Inductance

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Learning Outcomes

In this chapter, you'll learn…

- how a time-varying current in one coil can induce an emf in a second, unconnected coil.
- how to relate the induced emf in a circuit to the rate of change of current in the same circuit.
- how to calculate the energy stored in a magnetic field.
- how to analyze circuits that include both a resistor and an inductor (coil).
- why electrical oscillations occur in circuits that include both an inductor and a capacitor.

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Introduction

- Many traffic lights change when a car rolls up to the intersection.
- This process works because a large coil is placed under the street, which carries a current that changes with time.



- The car contains conducting material, so when it is near the coil, electric currents are induced in the car, which in turn induce an emf in the buried coil.
- We'll learn about this and other forms of inductance.

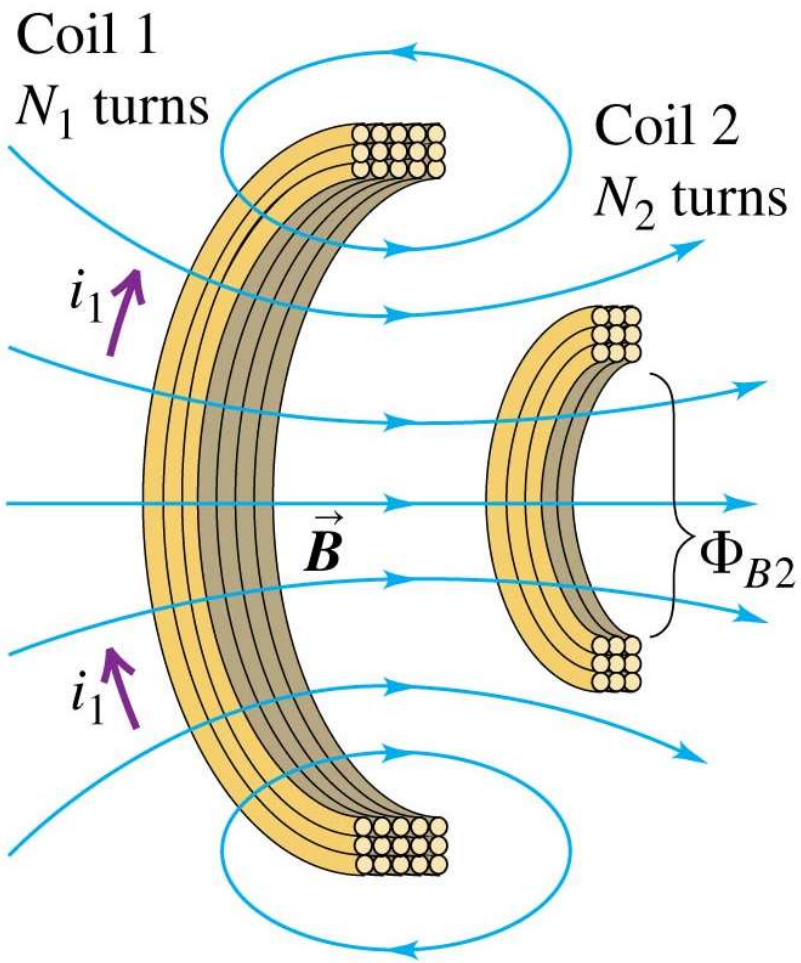
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Mutual Inductance (1 of 3)

- Consider two neighboring coils of wire, as shown.
- If the current in coil 1 changes, this induces an emf in coil 2, and vice versa.
- The proportionality constant for this pair of coils is called the **mutual inductance**, *M*.

$$\varepsilon_2 = -N_2 \frac{d\Phi_{B2}}{dt}$$

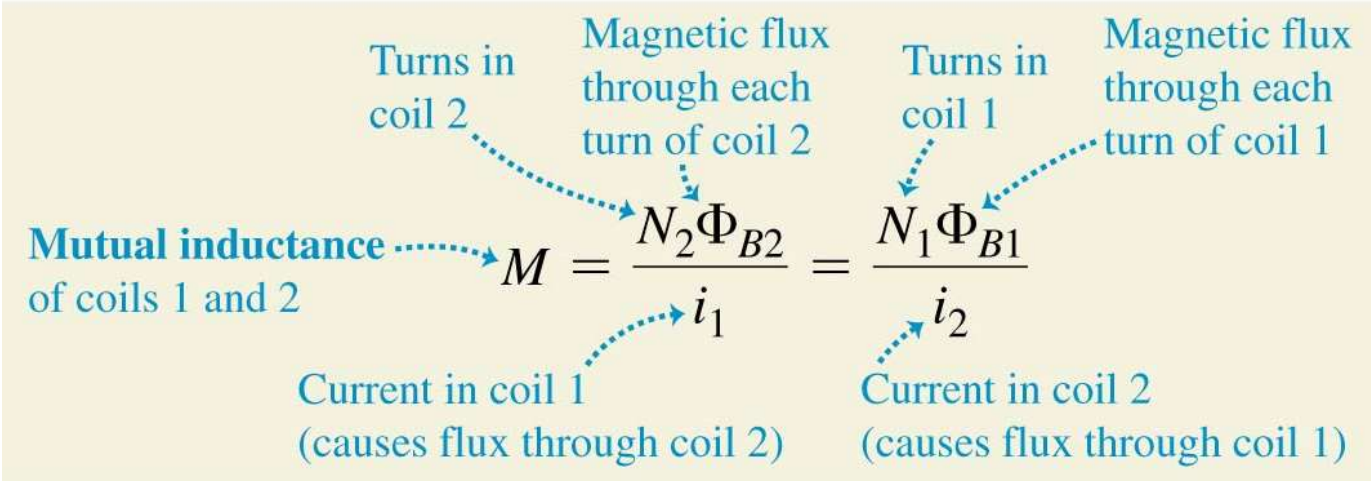


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Mutual Inductance (2 of 3)

- The mutual inductance M is:



- The SI unit of mutual inductance is called the henry (1 H), in honor of the American physicist Joseph Henry.

$$1\ \text{H} = 1\ \text{Wb/A} = 1\ \text{V}\cdot\text{s/A} = 1\Omega\cdot\text{s} = 1\ \text{J/A}^2$$

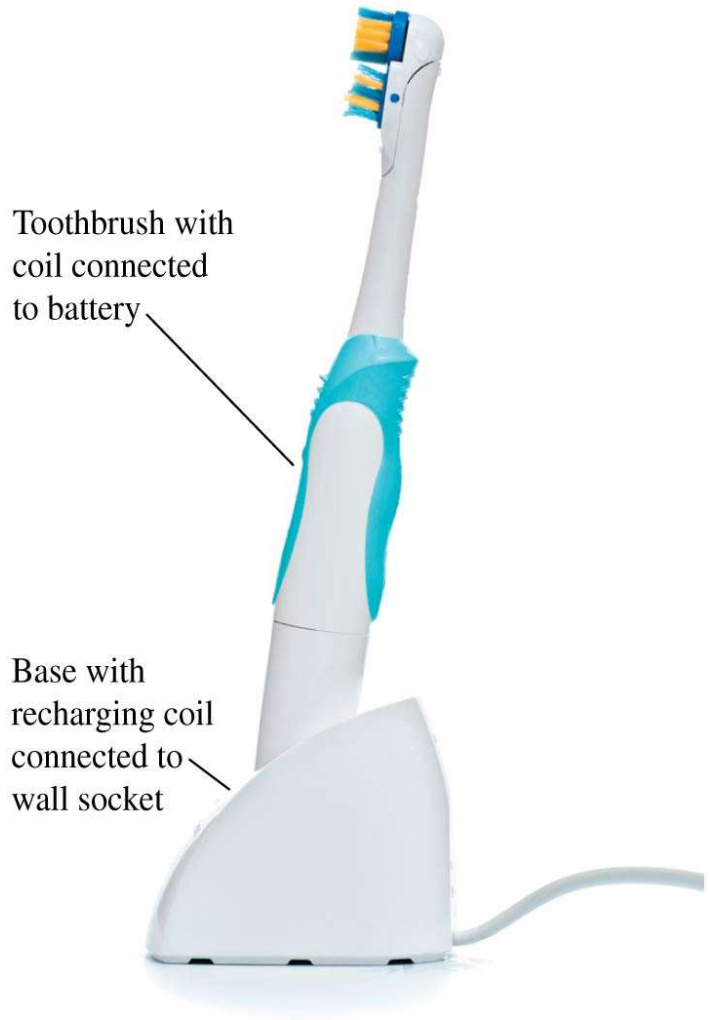
- Video Tutor Solution: Example 30.1

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Mutual Inductance (3 of 3)

- This electric toothbrush makes use of mutual inductance.
- The base contains a coil that is supplied with alternating current from a wall socket.
- Even though there is no direct electrical contact between the base and the toothbrush, this varying current induces an emf in a coil within the toothbrush itself, recharging the toothbrush battery.



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Self-Inductance

- Any circuit with a coil that carries a varying current has a **self-induced emf**.
- We define the self-inductance L of the circuit as:

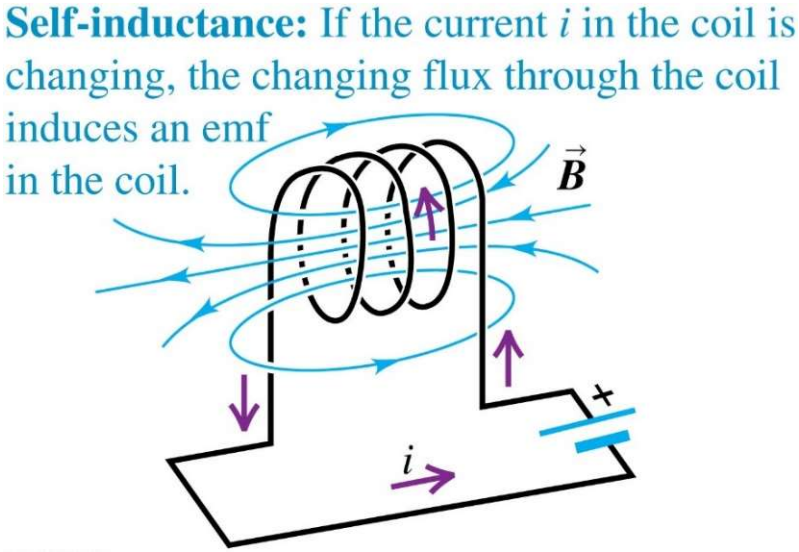
Self-inductance (or **inductance**) of a coil

$$L = \frac{N\Phi_B}{i}$$

N: Number of turns in coil

Φ_B: Flux due to current through each turn of coil

i: Current in coil



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Inductors and Lightning Strikes

- If lightning strikes part of an electrical power transmission system, it causes a sudden spike in voltage that can damage the components of the system.
- To minimize these effects, large **inductors** are incorporated into the transmission system.
- These use the principle that an inductor opposes and suppresses any rapid changes in the current.



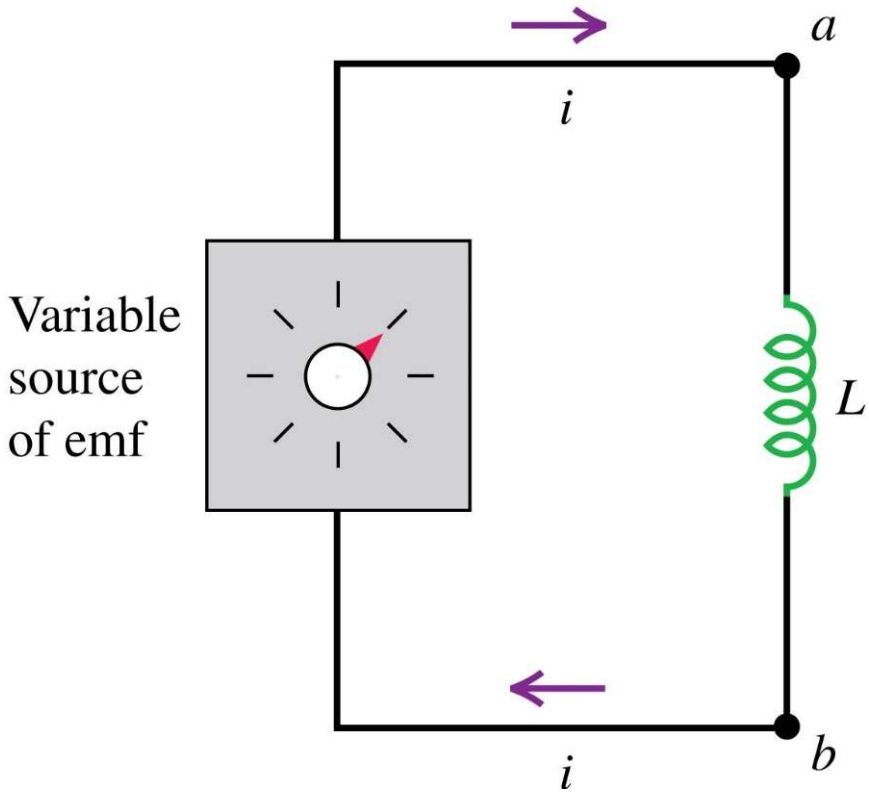
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Inductors as Circuit Elements

- In the circuit shown, the box enables us to control the current i in the circuit.
- The potential difference between the terminals of the inductor L is:

$$V_{ab} = V_a - V_b = L\frac{di}{dt}$$

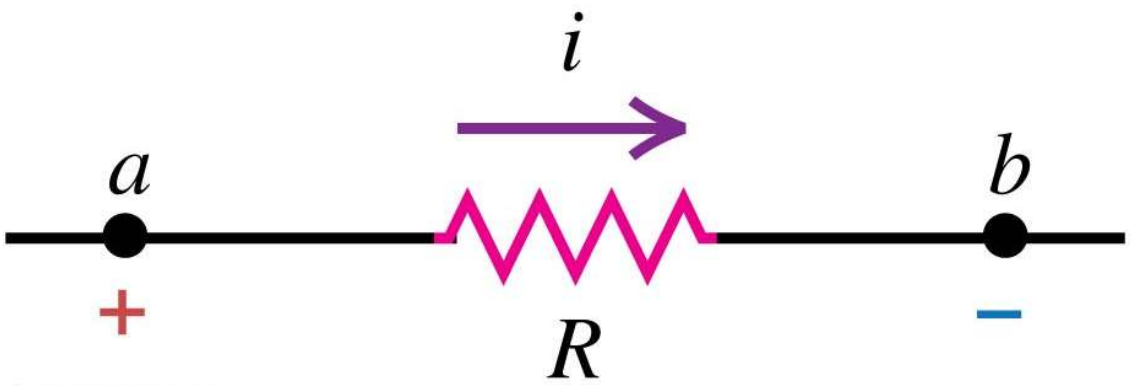


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Potential across a Resistor

- The potential difference across a resistor depends on the current.
- When you have a resistor with current *i* flowing from *a* to *b*, the potential **drops** from *a* to *b*.



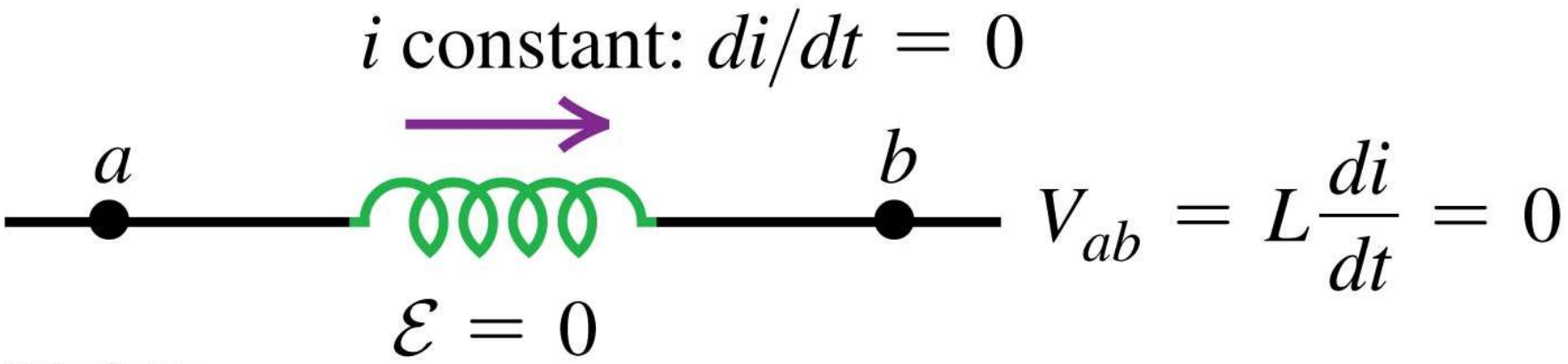
$$V_{ab} = iR > 0$$

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Potential across an Inductor with Constant Current

- The potential difference across an inductor depends on the rate of change of the current.
- When you have an inductor with **constant** current *i* flowing from *a* to *b*, there is **no** potential difference.

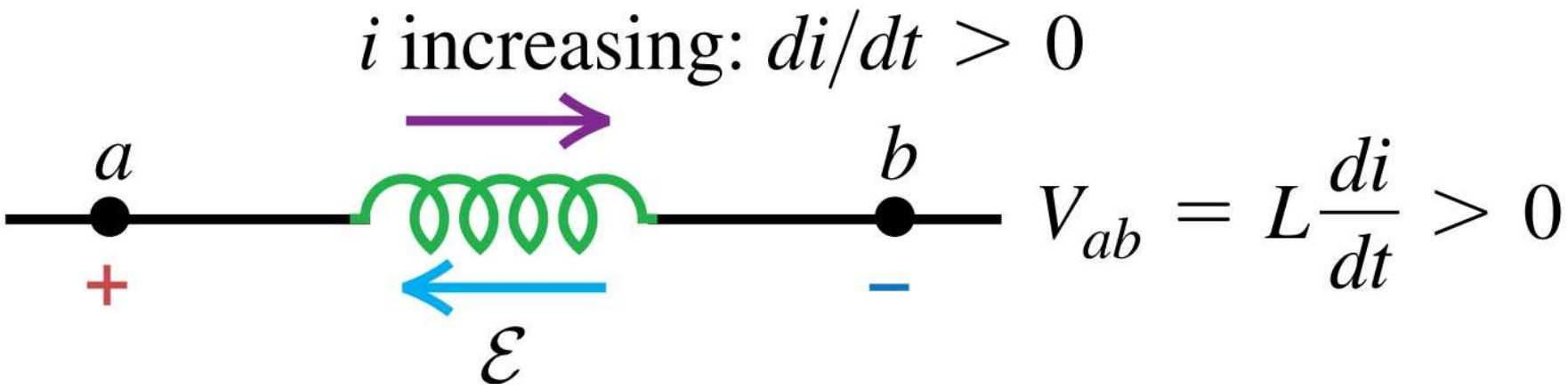


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Potential across an Inductor with Increasing Current

- The potential difference across an inductor depends on the rate of change of the current.
- When you have an inductor with **increasing** current *i* flowing from *a* to *b*, the potential **drops** from *a* to *b*.

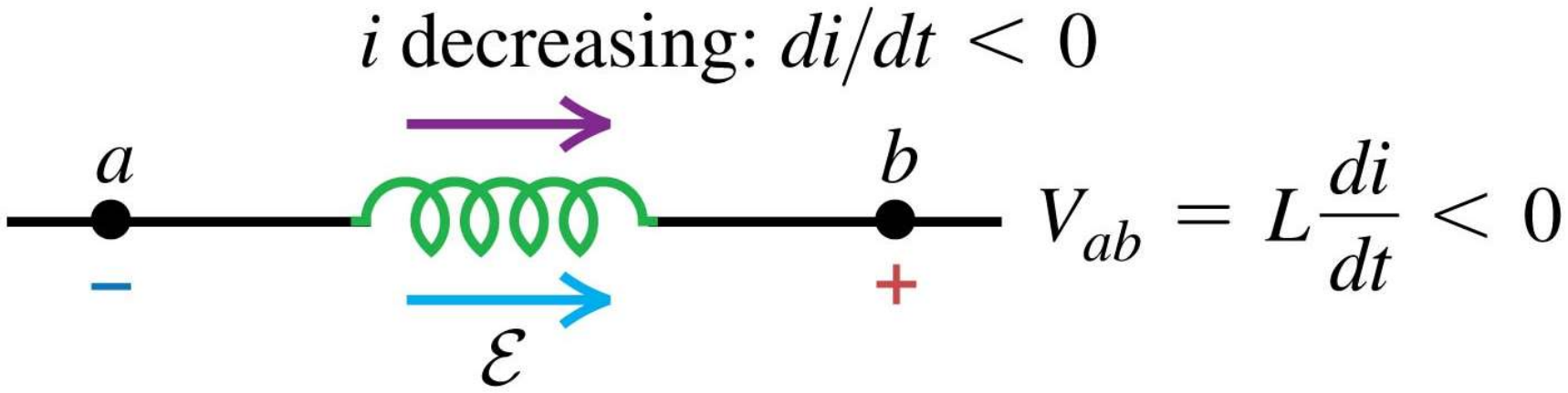


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Potential across an Inductor with Decreasing Current

- The potential difference across an inductor depends on the rate of change of the current.
- When you have an inductor with **decreasing** current *i* flowing from *a* to *b*, the potential **increases** from *a* to *b*.



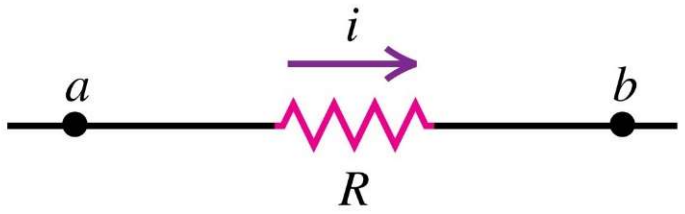
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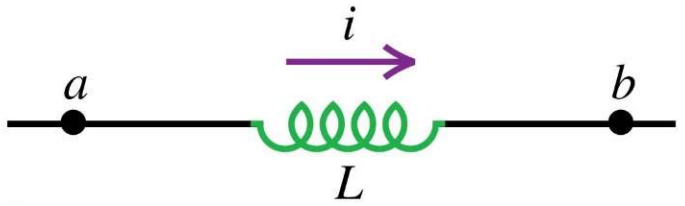
Magnetic Field Energy

- A resistor is a device in which energy is irrecoverably **dissipated**.
- By contrast, energy stored in a current-carrying inductor can be recovered when the current decreases to zero.

Resistor with current i: energy is *dissipated*.



Inductor with current i: energy is *stored*.



Energy stored in an inductor

$$U = L\int_0^I i\,di = \tfrac{1}{2}LI^2$$

Inductance

Final current

Integral from initial (zero) value of instantaneous current to final value

- Video Tutor Solution: Example 30.5

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Magnetic Energy Density

- The energy in an inductor is actually stored in the magnetic field of the coil, just as the energy of a capacitor is stored in the electric field between its plates.
- In a vacuum, the energy per unit volume, or **magnetic energy density**, is:

Magnetic energy density in vacuum

$$u = \frac{B^2}{2\mu_0}$$

(B^2: Magnetic-field magnitude; μ_0: Magnetic constant)

- When the magnetic field is located within a material with (constant) magnetic permeability $\mu = K_m \mu_0$, we replace μ_0 by μ in the above equation:

Magnetic energy density in a material

$$u = \frac{B^2}{2\mu}$$

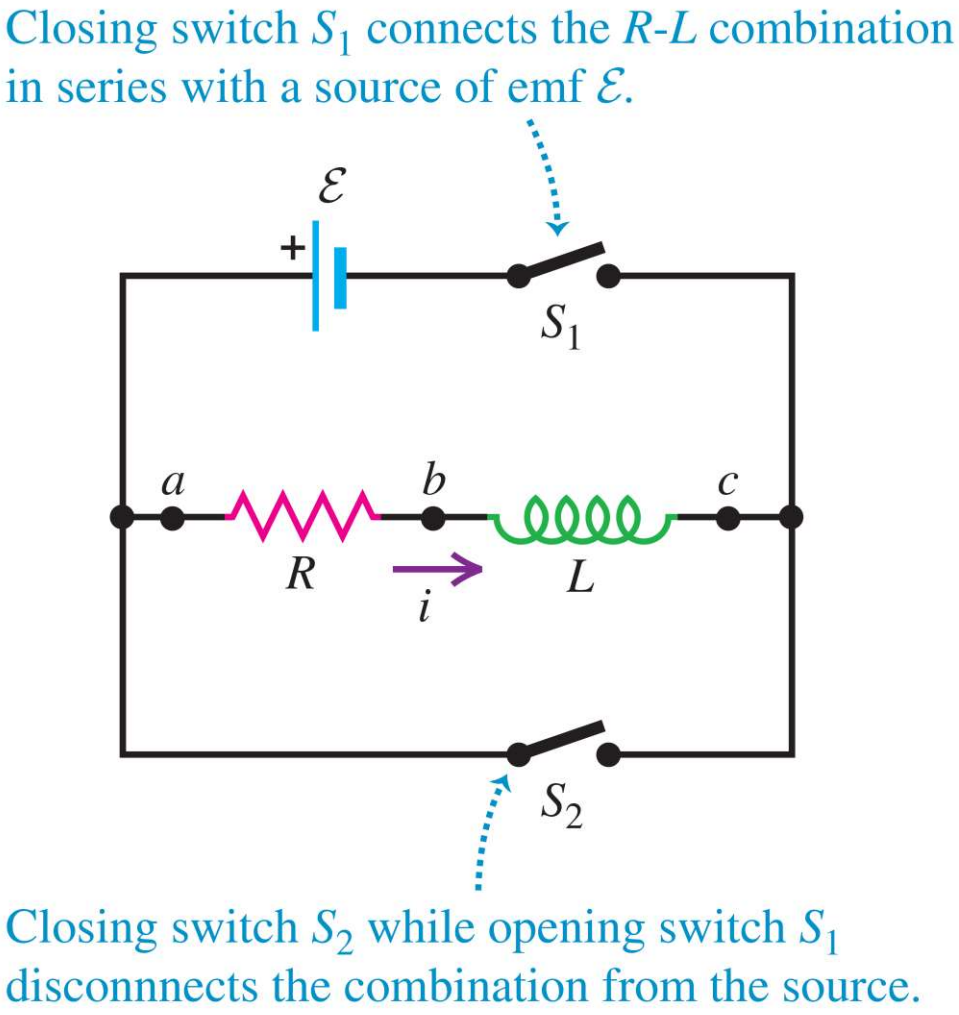
(B^2: Magnetic-field magnitude; μ: Permeability of material)

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The *R-L* Circuit

- An *R-L* **circuit** contains a resistor and inductor and possibly an emf source.
- Shown is a typical *R-L* circuit.

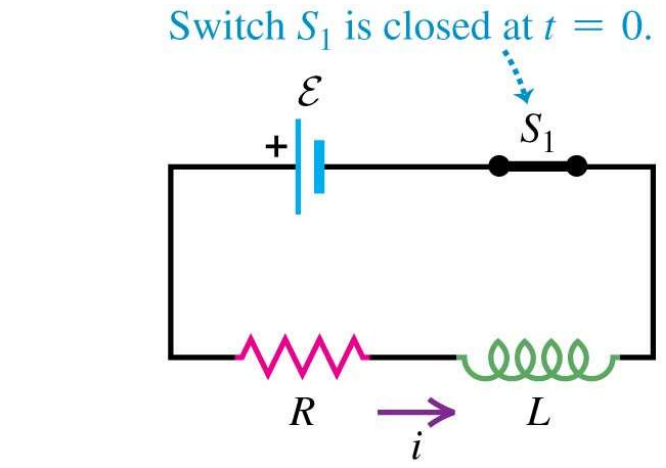


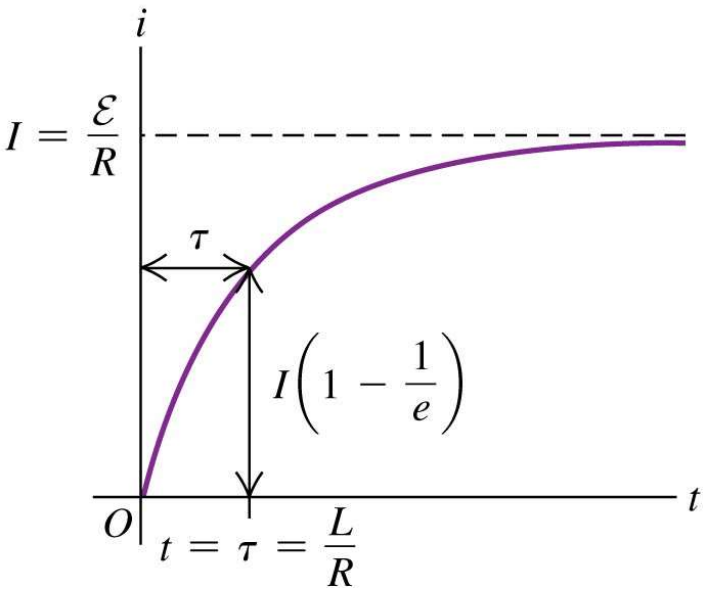
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Current Growth in an *R-L* Circuit

- Suppose that at some initial time $t = 0$ we close switch S_1.
- The current cannot change suddenly from zero to some final value.
- As the current increases, the rate of increase of current given becomes smaller and smaller.
- This means that the current approaches a final, steady-state value I.
- The **time constant** for the circuit is $\tau = \frac{L}{R}$.





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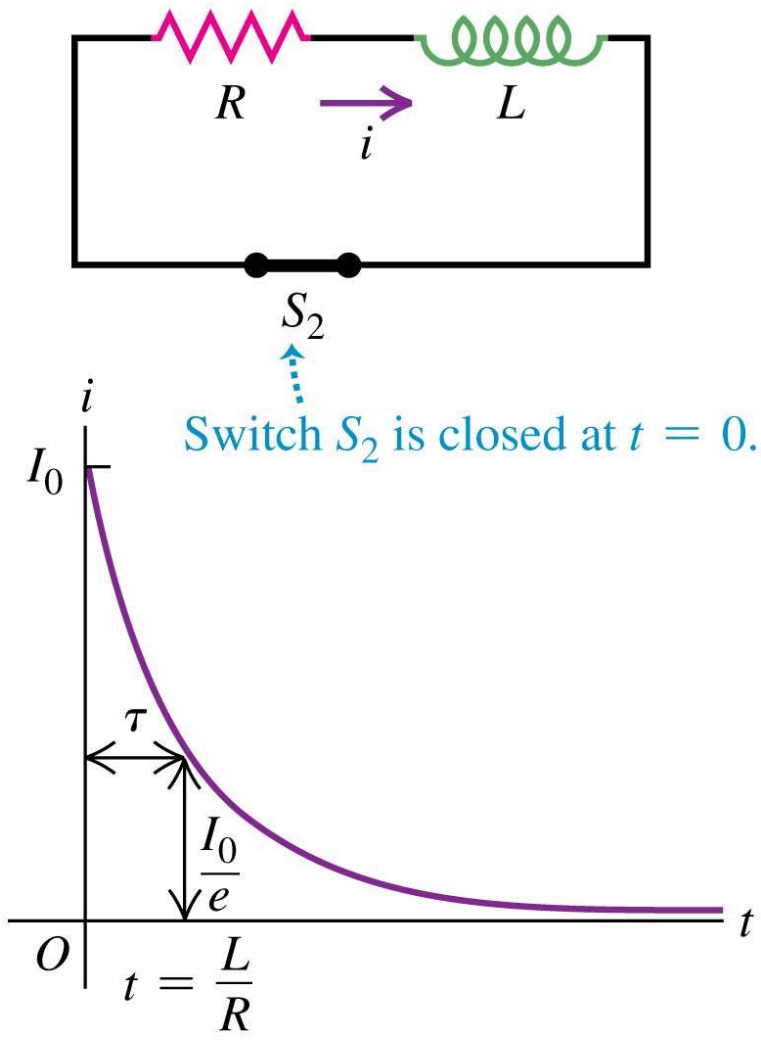
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Current Decay in an *R-L* Circuit

- Suppose there is an initial current I_0 running through the resistor and inductor shown.
- At time $t = 0$ we close the switch S_2, bypassing the battery (not shown).
- The energy stored in the magnetic field of the inductor provides the energy needed to maintain a decaying current.
- The **time constant** for the exponential decay of the current is

$$\tau = \frac{L}{R}.$$

- Video Tutor Solution: Example 30.7



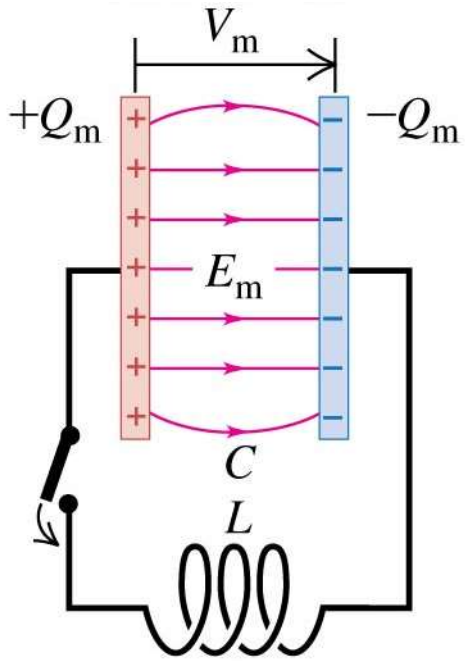
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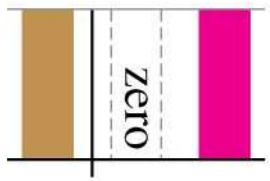
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The *L-C* Circuit: Oscillation (1 of 4)

Step 1:

- An **L-C circuit** contains an inductor and a capacitor and is an **oscillating** circuit.
- We will show the four main steps of the oscillation cycle on this and the next three slides (see Figure 30.14 of your text).
- In the *L-C* circuit shown we charge the capacitor to a potential difference V_m and initial charge $Q_m = CV_m$ on its left-hand plate and then close the switch.
- The capacitor is fully charged, the current is zero, and the circuit's energy is all stored in the electric field.





$E = U_B + U_E$

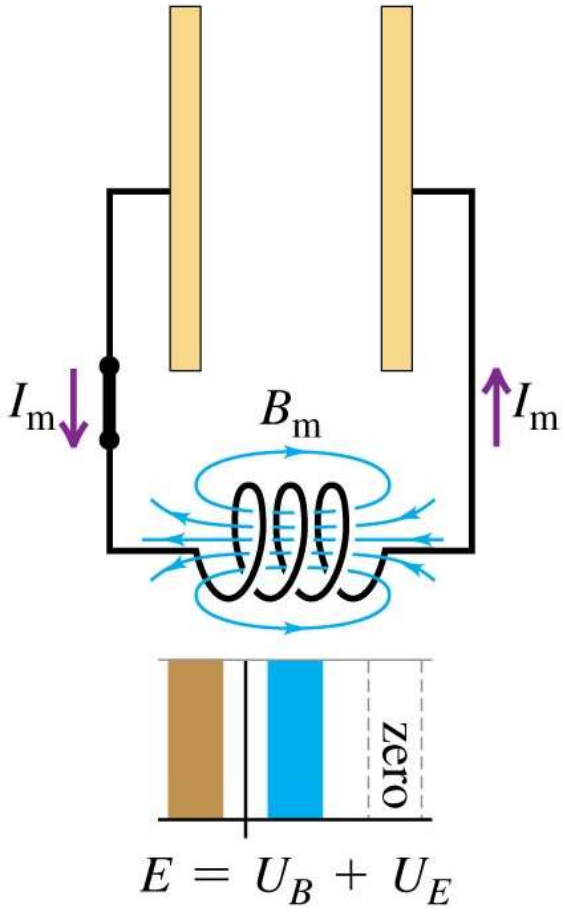
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The *L-C* Circuit: Oscillation (2 of 4)

Step 2:

- The capacitor discharges through the inductor.
- As the capacitor discharges, the current increases, but the rate of change of current decreases.
- When the capacitor potential becomes zero, the induced emf is also zero, and the current has leveled off at its maximum value I_m.
- Shown is this situation: The capacitor has completely discharged, the current is maximal, and the circuit's energy is all stored in the magnetic field.



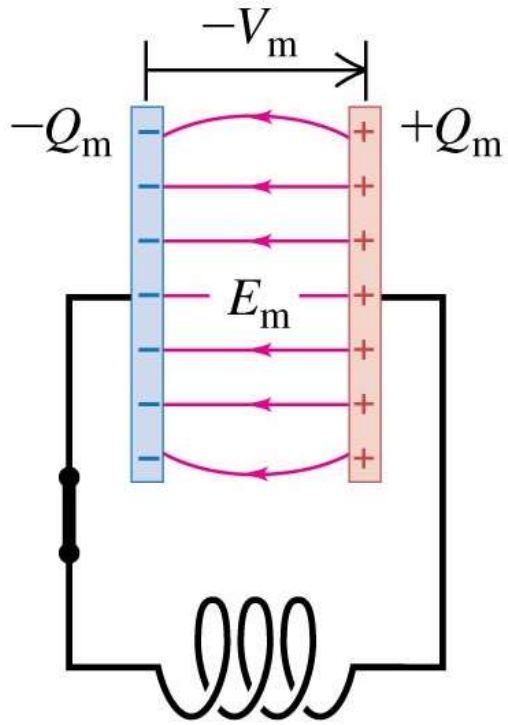
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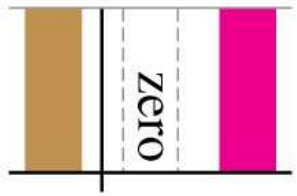
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The *L-C* Circuit: Oscillation (3 of 4)

Step 3:

- Although the capacitor was completely discharged in step 2, the current persists, and the capacitor begins to charge with polarity opposite to that in the initial state.
- Eventually, the current and the magnetic field reach zero, and the capacitor has been charged in the sense opposite to its initial polarity.
- Shown is this situation: The capacitor is fully charged, the current is zero, and the circuit's energy is all stored in the electric field.





$E = U_B + U_E$

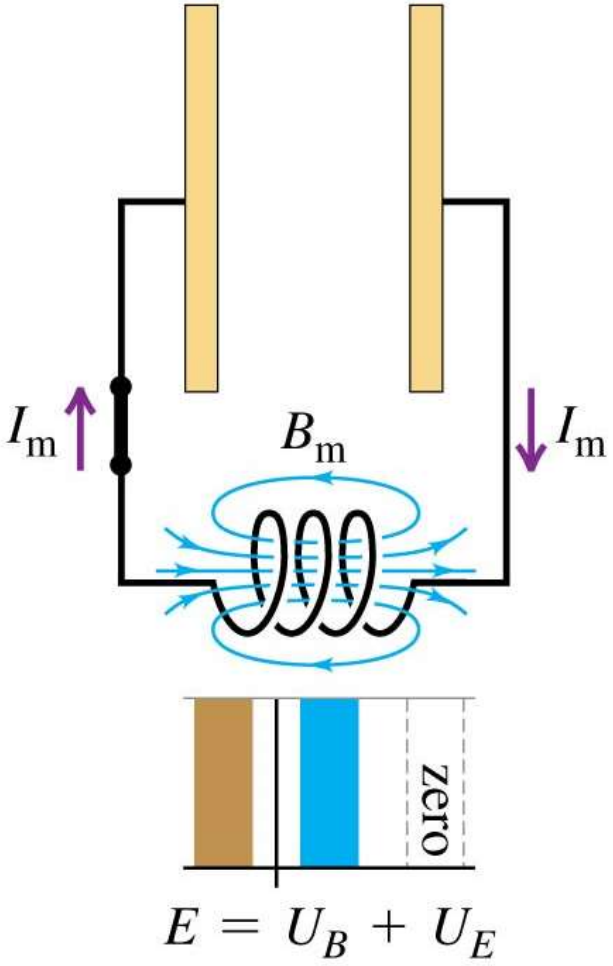
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The *L-C* Circuit: Oscillation (4 of 4)

Step 4:

- The process now repeats in the reverse direction; a little later, the capacitor has again discharged, and there is a current in the inductor in the opposite direction.
- Shown is this situation: The capacitor has completely discharged, the current is maximal, and the circuit's energy is all stored in the magnetic field.
- Still later, the capacitor charge returns to its original value, and the whole process repeats (return to step 1).



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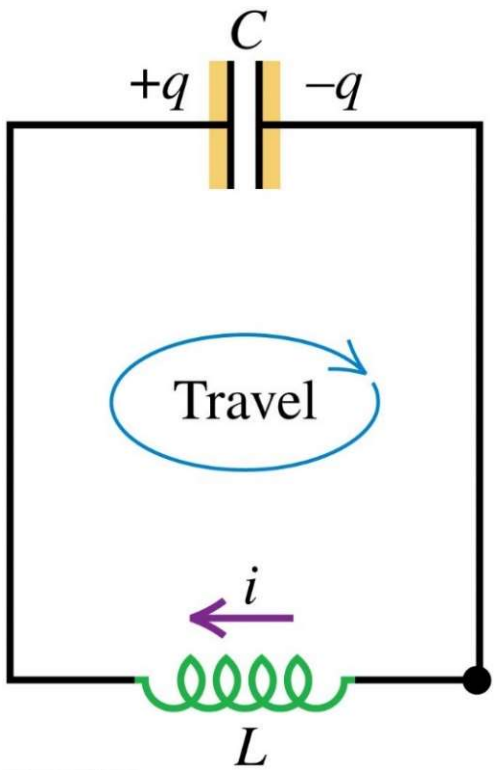
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Electrical Oscillations in an *L-C* Circuit

- We can apply Kirchhoff's loop rule to the circuit shown.
- This leads to an equation with the same form as that for simple harmonic motion studied in Chapter 14.
- The charge on the capacitor and current through the circuit are functions of time:

$$q = Q\cos(\omega t + \phi)$$

$$i = -\omega Q\sin(\omega t + \phi)$$



Angular frequency of oscillation in an *L-C* circuit

$$\omega = \sqrt{\frac{1}{LC}}$$

Capacitance

Inductance

- Video Tutor Solution: Example 30.8

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Electrical and Mechanical Oscillations: Analogies

Mass-Spring System

Kinetic energy $= \frac{1}{2}mv_x^2$

Potential energy $= \frac{1}{2}kx^2$

$$\frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$

$$v_x = \pm\sqrt{\frac{k}{m}}\sqrt{A^2 - x^2}$$

$$v_x = \frac{dx}{dt}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$x = A\cos(\omega t + \phi)$$

Inductor-Capacitor Circuit

Magnetic energy $= \frac{1}{2}Li^2$

Electrical energy $= \frac{q^2}{2C}$

$$\frac{1}{2}Li^2 + \frac{q^2}{2C} = \frac{Q^2}{2C}$$

$$i = \pm\sqrt{\frac{1}{LC}}\sqrt{Q^2 - q^2}$$

$$i = \frac{dq}{dt}$$

$$\omega = \sqrt{\frac{1}{LC}}$$

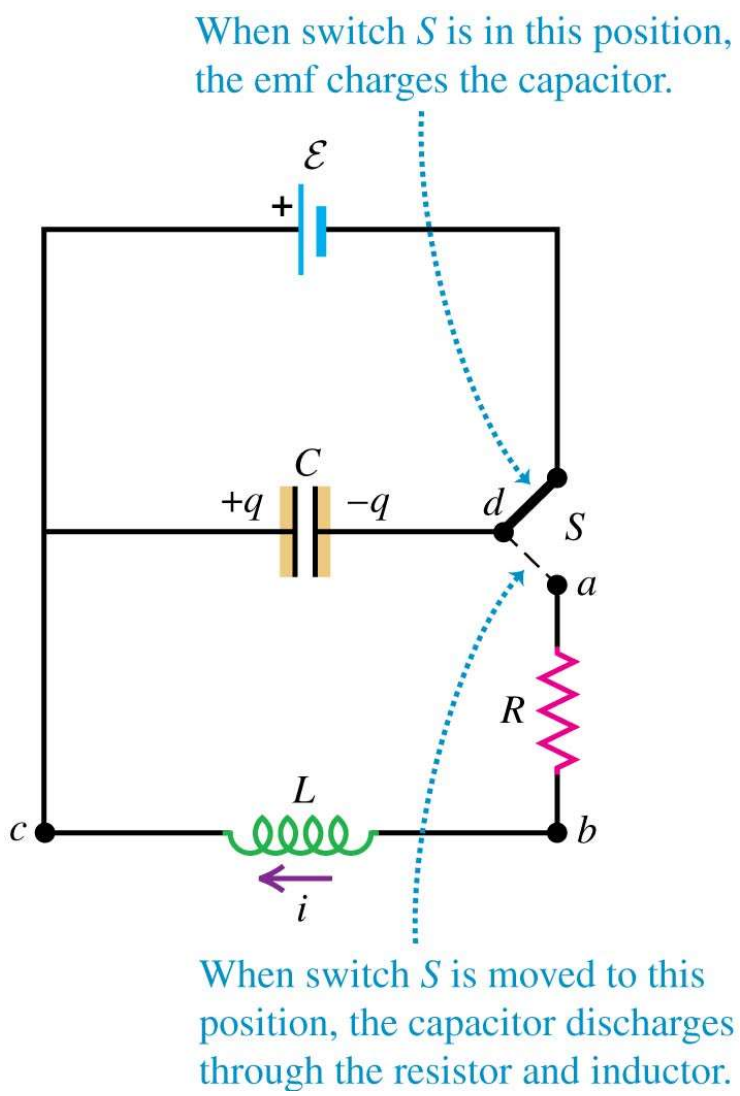
$$q = Q\cos(\omega t + \phi)$$

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The *L-R-C* Series Circuit (1 of 2)

- Consider the circuit shown.
- The emf source charges the capacitor initially.
- When the switch is moved to the lower position, we have an inductor with inductance *L* and a resistor of resistance *R* connected in series across the terminals of a charged capacitor, forming an ***L-R-C* series circuit**.

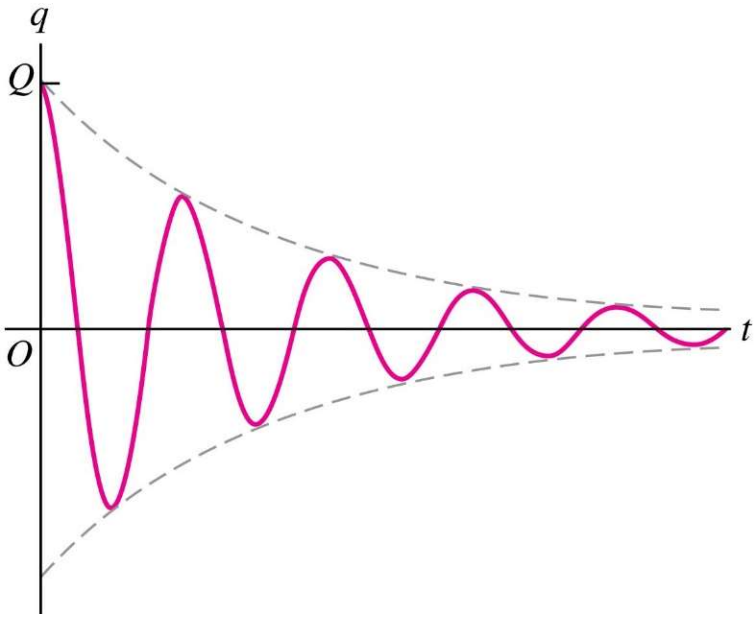


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The *L-R-C* Series Circuit (2 of 2)

- An *L-R-C* circuit exhibits **damped harmonic motion** if the resistance is not too large.
- The charge as a function of time is sinusoidal oscillation with an exponentially decaying amplitude, and angular frequency:



Angular frequency of underdamped oscillations in an *L-R-C* series circuit

$$\omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

R^2: Resistance; $4L^2$: Inductance; LC: Inductance, Capacitance

- Video Tutor Solution: Example 30.10

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End of chapter

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